

Figure 3: Data science workflow (Image credit: googleimages.com)

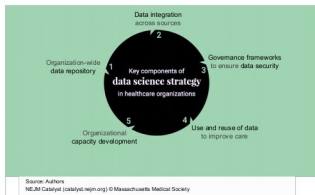


Figure 4: Different components of data science in a healthcare organisation (Image credit: googleimages.com)

What is machine learning?

Machine learning is a part of computer science that gives any system the ability to learn on its own without being programmed. It makes a machine learn in the same way as human beings learn by themselves. Just as we learn any system on the basis of our experience and the knowledge gained after analysing the system, even machines can analyse and study the system's behaviour or its output data and learn how to take decisions on that basis. This is the backbone of artificial intelligence. It makes machines get into a self-learning mode without any explicit programming. When the machine is fed with new data, it learns, grows and changes by itself.

Machine learning has evolved from the concept of pattern recognition and computational learning theory in artificial intelligence. It explores the study and construction of different algorithms that can learn from data and make predictions on them. These algorithms do not follow the static program instructions, but make data-driven predictions or decisions by building a model from some sample inputs.

There are three types of machine learning, differentiated on the basis of the learning signal available to any learning system.

1. **Supervised learning:** In this type of learning, the machine is presented with few example inputs and also their desired outputs, which are given by a teacher. The goal is to learn a general rule that maps inputs to outputs.

2. **Unsupervised learning:** This is a type of machine learning in which no labels are given to the learning algorithm, leaving it to find the structure in its input on its own.
3. **Reinforcement learning:** Under this learning system, a computer program actually interacts with a dynamic environment for which it must perform a specific goal (for example, driving a vehicle or playing a game against an opponent).

How is machine learning related to data science?

Machine learning is very closely related to (and sometimes overlaps with) data science or computational statistics, as both focus on making predictions with the help of machines or computers. It has strong ties with mathematical optimisation, which provides different methods and theories to optimise learning systems. Machine learning is often combined with data science, with the latter actually focusing more on exploratory data analysis, and this is known as unsupervised learning.

If we talk specifically about the field of data science, machine learning is used to devise various complex models and algorithms that lend themselves to prediction. This is also known as predictive analytics. All these analytical models allow data scientists, researchers, engineers and analysts to produce reliable and repeatable decisions in order to uncover various hidden insights by learning from the historical relationships and trends in the large sets of data.

Data analysis has been traditionally characterised by the trial and error approach, and we all know that this becomes impossible to use when there are large and heterogeneous sets of data to be analysed. The availability of large data is directly proportional to the difficulty of developing new predictive models that work accurately. All the traditional statistical solutions work for static analysis, which is limited to the analysis of samples frozen in time. Machine learning has emerged as a solution to all this chaos, proposing different clever alternatives to analyse huge volumes of data. It is able to produce accurate results and analyses by developing various efficient and fast working algorithms for the real-time processing of data.

Some applications of machine learning

Machine learning has been implemented in a number of applications. Some of them are:

1. Google's self-driving car
2. Online recommendation engines such as friend recommendation on Facebook
3. Various offer recommendations from Amazon
4. Cyber fraud detection
5. Optical character recognition (OCR)

The role of machine learning in data science

1. Machine learning helps to analyse large chunks of data easily and hence eases the work of data scientists in an automated process.